

Materials and Methods

Structure-corrected two-locus association for intrinsic incompatibilities between unlinked LD-reduced units

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Draft prepared from the current /doc pipeline documentation

This is a self-contained working draft for integration into the full Supplementary Materials, describing the intrinsic (Dobzhansky–Muller) arm that complements the ancestry-sorting and climate analyses documented separately. Blue boxes identify items that remain open; in particular the neutral, recombination-matched null (Section S7) is still required to convert the descriptive leads into excess-over-neutral statements.

S1 Scope and analytical principles

We screened for residual associations between unlinked genomic regions that could identify candidate two-locus incompatibilities. The analysis used the LD-reduced units, expected multi-locus genotype (eMLG) consensus dosages, and parental allele-frequency gate described in the companion methods.

Associations between unlinked regions can arise through epistatic selection, but also through shared ancestry, relatedness among nestmates, or technical duplication. We therefore used a mixed model to reduce covariance associated with ancestry and relatedness, removed likely duplicate mappings, and required associations to be robust to omission of any one population. Because the focal regions formed a targeted rather than representative genome-wide set, the resulting associations are interpreted as exploratory candidates. The hybrid populations are also not assumed to be fully independent evolutionary replicates.

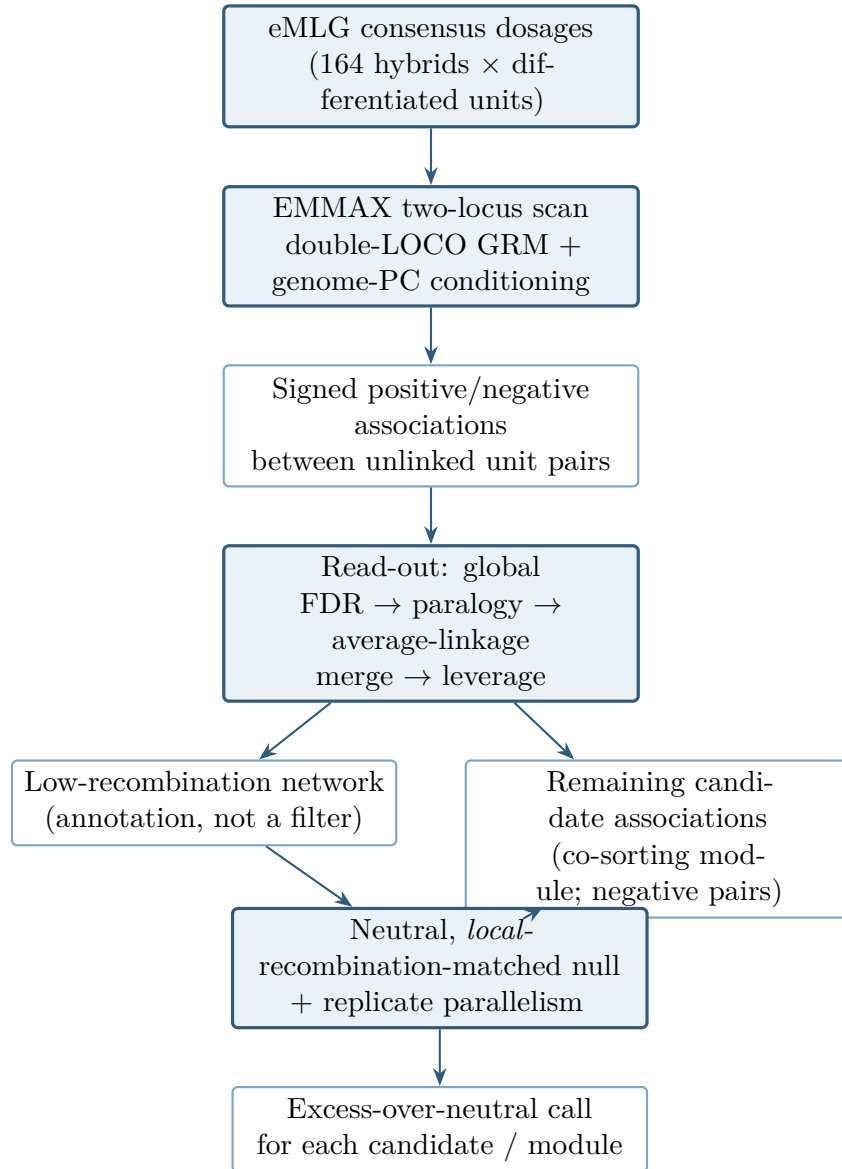


Figure 1: **Analytical workflow for the intrinsic arm.** A single structure-corrected scan is summarized by one procedure into a low-recombination co-ancestry network and a short candidate list. Low recombination is annotated, never used to discard loci; every unit is carried into the neutral, *local*-recombination-matched null that provides the final call.

S2 Units, orientation, and the unlinked definition

The analytical unit is the eMLG consensus dosage of an LD cluster (continuous, on an *F. aquilonia*-oriented 0–2 scale; no hard-calling), restricted to the parent-differentiated units used as informative markers throughout. Consensus orientation and the parental gate (pooled parental MAF ≥ 0.15) follow the companion methods, so that a unit’s dosage and its diagnostic index (DI) are on consistent scales.

Two units were treated as *unlinked* when they lay on different chromosomes, or on the same chromosome more than $\text{LINK}_{\text{cM}} = 10$ cM apart on the genetic map. At the recombination rates of this genome, 10 cM corresponds to approximately 99% decay of admixture LD over the estimated ~ 50 generations since contact. Per-unit map positions were calculated as the median positions of their member markers. The 10-cM criterion is a deliberately conservative threshold for this association screen and is distinct from the 0.5-cM merge limit used to construct the LD-reduced units.

S3 Structure-corrected two-locus association (EMMAX)

S3.1 Rationale and model

To ask whether two units are associated *beyond* what shared ancestry and relatedness predict, we treat one unit’s consensus dosage as a quantitative trait and test its association against every other differentiated unit in a linear mixed model,

$$y_i = \mu + \beta x_j + u + \varepsilon, \quad u \sim N(0, \sigma_g^2 K), \quad \varepsilon \sim N(0, \sigma_e^2 I), \quad (1)$$

where y_i is the focal unit’s dosage across the 164 individuals, x_j a candidate partner, and K a genomic-relationship matrix (EMMAX; Li, Kempainen, Rastas & Merilä). This is a genome scan in which the “phenotype” is one unit and the “markers” are all the others; variance components are estimated once under the null and reused for every partner.

S3.2 Controlling for ancestry and relatedness

The genomic-relationship matrix captures broad genetic structure and relatedness among individuals. In addition, each focal dosage was adjusted for the first ten genetic principal components. Three model choices were used:

1. **Relationship matrix.** K was built from differentiated units because these units capture the ancestry variation most relevant to the scan. A matrix based on a neutral background produced residual test-statistic inflation.
2. **Double leave-one-chromosome-out.** Because the scan is symmetric, both the focal and the partner chromosome are dropped from K , so neither tested unit’s own region deflates its association. This keeps the test calibrated ($\lambda \approx 0.98$ –1.05; Fig. 2).
3. **Genetic-PC adjustment.** Each focal dosage was adjusted for the first ten principal components calculated from the differentiated eMLGs. This reduced residual associations with secondary axes of genetic structure.

Each association is signed by its structure-corrected generalised-least-squares coefficient (positive or coupling, +; negative or repulsion, –). The targeted scan scanned 51 focal units (bidirectional

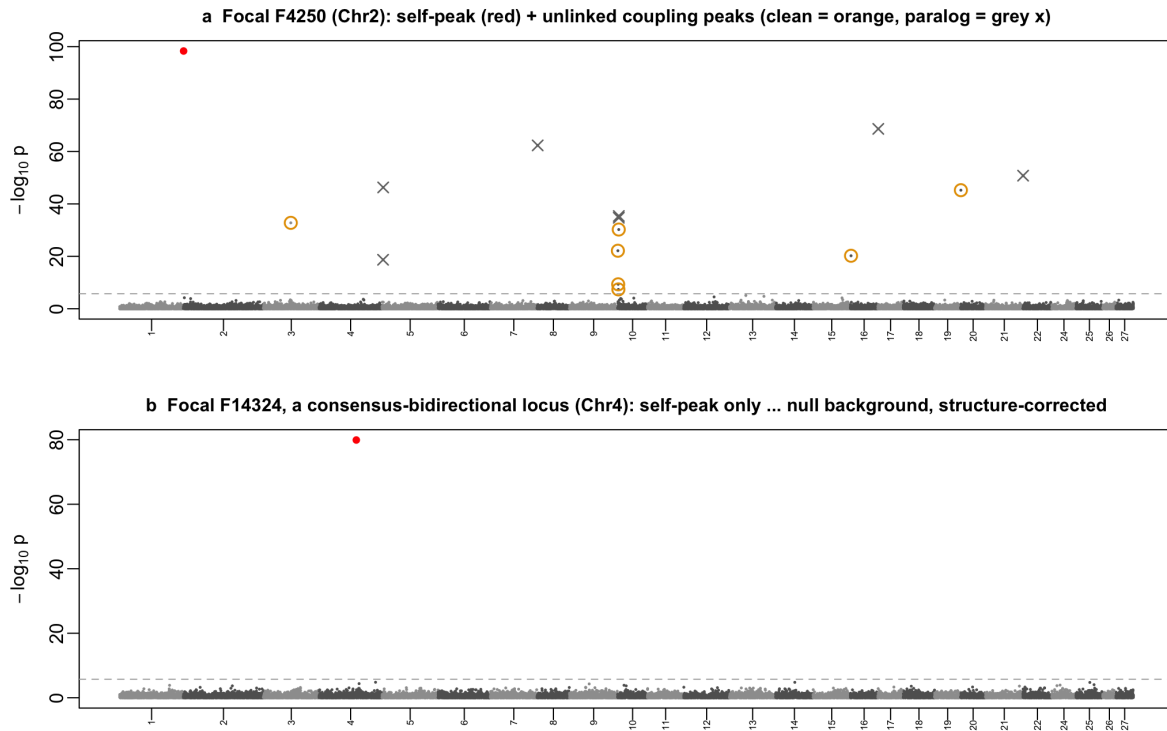


Figure 2: **Calibration and read-out of the structure-corrected scan.** Diagnostic summary of the double-LOCO, genome-PC-conditioned EMMAX scan: quantile–quantile calibration of the association statistic ($\lambda \approx 0.98$ – 1.05), an example focal Manhattan with its self-locus and linked block removed, and the signed positive and negative associations. The scan is calibrated and its strongest associations are cross-chromosome, motivating the paralogy filter of Section S4.

candidates, extreme among-population-covariance units, and controls including a random-focal set for calibration).

False-discovery rates therefore refer to all comparisons in this targeted screen, not to an unbiased genome-wide sample of focal regions.

S4 Cross-chromosome paralogy

Duplicate sequence or incorrect mapping can produce nearly identical genotypes at different assembly locations. Such pairs cannot be removed using genetic distance. We therefore calculated the correlation between each pair’s consensus dosages within each population and used the median absolute correlation across populations. Pairs with median $|r| > \text{PARALOGY}_r = 0.9$ were flagged as potential duplicate or mismapped regions. Genotype concordance and excess observed heterozygosity were used as supporting diagnostics (Fig. 3). These criteria identify candidate technical artefacts rather than proving paralogy.

S5 Filtering and network summarization

The scan was read out by one procedure rather than per-focal thresholds, so that the structural and pair-specific components separate under a single criterion.

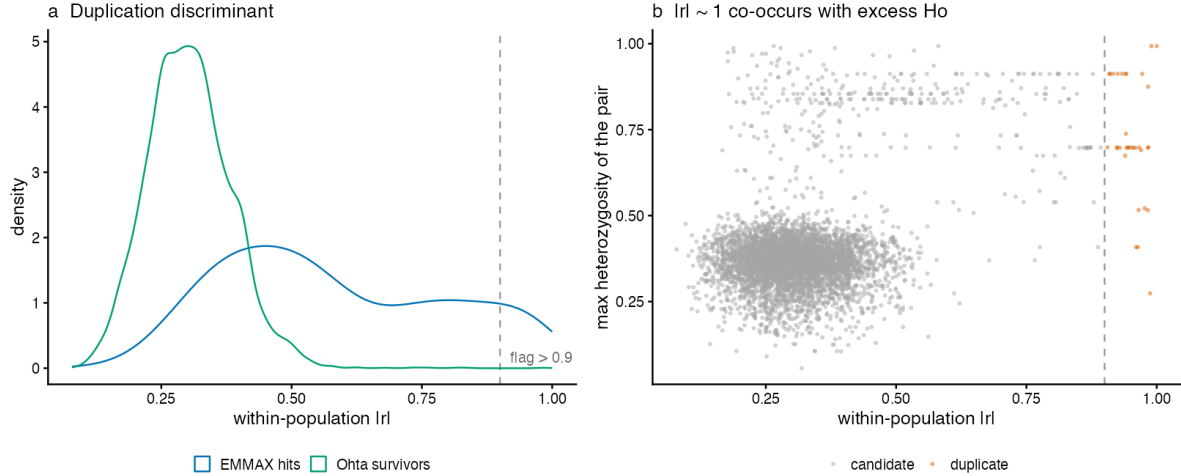


Figure 3: **The paralogy discriminant.** Distribution of median within-population $|r|$ for unlinked pairs, with the 0.9 flag; the corroborating rise in genotype concordance and per-unit excess heterozygosity toward $|r| = 1$; and the separation of flagged (technical) from retained (candidate) pairs. Distance filtering is orthogonal to this signal-based filter, so both are applied.

1. **False-discovery control.** All unlinked focal-partner tests (1.38×10^6) were pooled and controlled at a global Benjamini-Hochberg $q < Q_{\text{FDR}} = 0.01$ across the targeted screen.
2. **Paralogy filter** (Section S4).
3. **Regional consolidation.** For this network only, correlated units on the same chromosome and within 10 cM were combined by average-linkage clustering ($|r| > \text{MERGE}_r = 0.5$). This prevented one extended region from being counted as several interacting loci.
4. **Leave-one-population-out robustness.** An edge was retained only if its among-population allele-frequency correlation survived leave-one-population-out analysis ($\min_k |r_{-k}| \geq \text{LEV}_{\text{LOO}} = 0.3$).

Cross-chromosome correlation *modules* were then defined by average-linkage clustering of the meta-node consensus *across* chromosomes, cut at $|r| > \text{MODULE}_r = 0.4$. Unlike the within-chromosome merge, this groups co-varying regions genome-wide. These modules were used as descriptive summaries and for genotype visualization.

S6 Recombination annotation, not a filter

Each meta-node was annotated by its recombination percentile relative to the differentiated background. Units below the tenth percentile ($\text{STRUCT}_{\text{pct}} = 0.10$) were labelled low-recombination regions. This was a descriptive annotation rather than an exclusion criterion. Low recombination can preserve long ancestry blocks and can also contribute to the maintenance of coadapted variation; consequently, it does not by itself distinguish neutral from selective explanations.

S7 Neutral comparison

The empirical scan cannot determine whether the retained associations exceed expectations under neutral admixture. This requires applying the same scan to simulations generated on

the empirical recombination map, with comparable ancestry, sampling, and population history. Observed pairs and modules can then be compared with simulated regions experiencing similar local recombination. Until that comparison is complete, the retained associations are treated as exploratory candidates rather than inferred incompatibilities.

S8 Statistical interpretation

The analyses are conditional on the sampled populations, quality filters, and targeted focal set. The mixed model reduces covariance associated with ancestry and relatedness but cannot guarantee that all structure has been removed. Low recombination is used as context rather than evidence for either neutrality or selection. No pair is called an incompatibility on the basis of this association screen alone; retained associations require comparison with the neutral simulations and independent biological scrutiny.

Table 1: Principal parameter settings for the intrinsic (two-locus) arm.

Analysis	Parameter	Value
Units	Analytical unit	eMLG consensus dosage (differentiated)
	Parental MAF gate	≥ 0.15
Unlinked	Genetic-distance rule	different chr, or same chr > 10 cM
EMMAX	Model	LMM, one unit as trait vs all others
	Genomic-relationship matrix K	VanRaden, differentiated units
	Leave-one-out	double (focal + partner chromosome)
	Genome-PC conditioning	top 10 PCs residualised from the focal
	Sign	structure-corrected GLS coefficient
Paralogy	Calibration	$\lambda \approx 0.98\text{--}1.05$
	Discriminant	median within-population $ r $
Read-out	Flag threshold	> 0.9
	Multiple testing	Benjamini–Hochberg $q < 0.01$ across targeted scan
	Regional consolidation	average linkage, $ r > 0.5, \leq 10$ cM
	Leverage	leave-one-population-out $ r \geq 0.3$

Analysis	Parameter	Value
	Modules	average linkage across chr, $ r > 0.4$
Annotation	Low-recombination label (not a filter)	recombination percentile < 0.10
Neutral comparison	Matching	local recombination (real map)
	Corroboration	replicate-population parallelism

References

- [1] Kang HM, Sul JH, Service SK, Zaitlen NA, Kong S-Y, Freimer NB, Sabatti C, Eskin E. Variance component model to account for sample structure in genome-wide association studies. *Nature Genetics*. 2010;42:348–354.
- [2] Nouhaud P, Martin SH, Portinha B, Sousa VC, Kulmuni J. Rapid and predictable genome evolution across three hybrid ant populations. *PLoS Biology*. 2022;20:e3001914. doi:10.1371/journal.pbio.3001914.